

[T R L] T R L

TRL documentation TRL - Transformers Reinforcement
documentation experience to get started TRL is a
of tools to train...

来源：<https://huggingface.co/docs/trl/index>

分类：09_模型微调与优化

TRL documentation

TRL - Transformers Reinforcement Learning

and get access to the augmented documentation exp

to get started

TRL is a full stack library where we provide a se
models with methods like Supervised Fine-Tuning (SFT),
(GRPO), Direct Preference Optimization (DPO), Rev

The library is integrated with transformers.

What's New

TRL v1: We released TRL v1 — a major milestone th
the blog post to learn more.

Taxonomy

Below is the current list of TRL trainers, organi
experimental).

Online methods

GRPOTrainer

GRPOTrainer

RLOOTrainer

RLOOTrainer

OnlineDPOTrainer

OnlineDPOTrainer

NashMDTrainer

NashMDTrainer

PPOTrainer

PPOTrainer

XPOTrainer

XPOTrainer

Reward modeling

RewardTrainer

RewardTrainer

PRMTrainer

PRMTrainer

Offline methods

SFTTrainer

SFTTrainer

DPOTrainer

DPOTrainer

BCOTrainer

BCOTrainer

CPOTrainer

CPOTrainer

KTOTrainer

KTOTrainer

ORPOTrainer

ORPOTrainer

Knowledge distillation

GKDTTrainer

GKDTTrainer

MiniLLMTrainer

MiniLLMTrainer

You can also explore TRL-related models, datasets organization.

Learn

Learn post-training with TRL and other libraries

Contents

The documentation is organized into the following

Getting Started: installation and quickstart guide

Conceptual Guides: dataset formats, training FAQ,

How-to Guides: reducing memory usage, speeding up

Integrations: DeepSpeed, Liger Kernel, PEFT, etc.

Examples: example overview, community tutorials,
API: trainers, utils, etc.

Blog posts

Published March 27, 2026

TRL v1: Post-Training Library That Holds When the

Published October 23, 2025

Building the Open Agent Ecosystem Together: Intro

Published on August 7, 2025

Vision Language Model Alignment in TRL

Published on June 3, 2025

NO GPU left behind: Unlocking Efficiency with Co-

Published on May 25, 2025

Liger GRPO meets TRL

Published on January 28, 2025

Open-R1: a fully open reproduction of DeepSeek-R1

Published on July 10, 2024

Preference Optimization for Vision Language Models

Published on June 12, 2024

Putting RL back in RLHF

Published on September 29, 2023

Finetune Stable Diffusion Models with DDPO via TF

Published on August 8, 2023

Fine-tune Llama 2 with DPO

Published on April 5, 2023

StackLLaMA: A hands-on guide to train LLaMA with

Published on March 9, 2023

Fine-tuning 20B LLMs with RLHF on a 24GB consumer

Published on December 9, 2022

Illustrating Reinforcement Learning from Human Feedback
Talks

Talk given on October 30, 2025

Fine tuning with TRL